Step 7: Analysis and Recommendations

Summary

Of the 7,043 customers, 5,174 (73.5%) stayed and 1,869 (26.5%) left. The initial KNN model (K = 5) achieved 76.44% test accuracy and flagged 43.58% of the customers who actually left. Among the settings tested, K = 11 had the highest overall accuracy (78.78%) and churn precision (66.23%), while K = 3 had the highest churn recall (45.19%). Customers who left tended to have shorter tenures and higher monthly charges. Even the setting with the highest recall missed more than half of the customers who left, so I would avoid relying on the model alone to decide who gets retention outreach.

Model Performance

What is the model's accuracy?

- The model accuracy is 82.80% on training data and 76.44% on test. The overall accuracy of 76% looks reasonable, but it is partly influenced by there being more customers who stayed than churned. The breakdown of the customers (stayed: 73.5%, left: 26.5% shows that a model that guessed ‘stayed’ for every customers of the dataset would already be right about 73.5% of the time.

  Regarding recall and precision: The model is better at identifying customers who stay than customers who leave. It correctly identified 88% of the customers who stayed, but only 44% of the customers who left.

[28] ✓ 0.1s Python

```
=== Model Performance ===

Training Accuracy: 0.8280 (82.80%)
Test Accuracy: 0.7644 (76.44%)

Test Precision: 0.5739
Test Recall: 0.4358

=== Confusion Matrix ===
                    Predicted
                    Stayed  Churned
Actual Stayed          914      121
Actual Churned         211      163

=== Classification Report ===
              precision    recall  f1-score   support

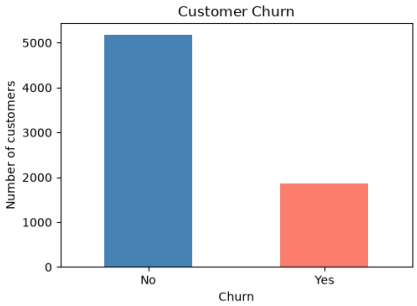
      Stayed       0.81      0.88      0.85      1035
     Churned       0.57      0.44      0.50       374

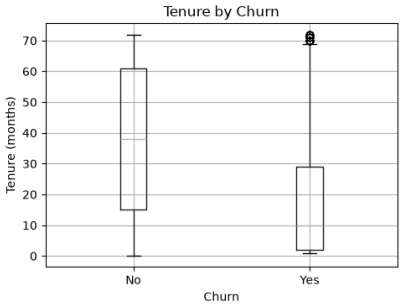
    accuracy                           0.76      1409
   macro avg       0.69      0.66      0.67      1409
weighted avg       0.75      0.76      0.75      1409
```

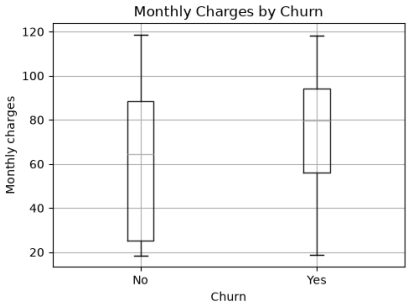
	K	Overall accuracy	Churn precision	Churn recall
0	1	70.83%	45.01%	44.65%
1	3	76.01%	55.96%	45.19%
2	5	76.44%	57.39%	43.58%
3	7	78.07%	62.26%	44.12%
4	9	78.50%	64.26%	42.78%
5	11	78.78%	66.23%	40.91%
6	15	78.71%	66.09%	40.64%

So our analysis shows that if we consider overall accuracy or churn precision to be our most important markers, then K11 is the K with the best value. If we consider churn recall: of the customers who actually cancelled, what is the percentage that actually cancelled.

What features seem most important? (You can explore this by looking at feature distributions)







- The box plot shows that the middle tenure for the customers who stayed was 38 months. The middle tenure for customers who left was only 10 months. Customers who left tended to be at the company for a shorter period of time
- Similarly if we look at the monthly charges by turn, the average (median) for those who stayed was around 65. For those who left, it was around 80.Customers who left tended to pay more per month. That gives the company a reason to look more closely at pricing, plan types, and what customers get for their money. The company can also evaluate whether there is a pattern where newer customers are getting a more pricy plan which makes them unhappy and leave or there are other factors at play

Business Recommendations

What would you recommend to the company based on your model?

The model is not ideal if our goal is to use this model to find out which customers are likely to leave or cancel and to device strategies to reach those customers. Just looking at the confusion matrix without even calculating any percentages we can see that the model caught only 163 people who actually left or cancelled and missed 211 who actually left.

If we use this model's flags as a call list, we would reach fewer than half of the customers who went on to leave.

”What are the limitations of your model?

One limitation is that even though K = 11 had the highest overall accuracy, it only flagged about 40 out of every 100 customers who actually left.

Because I didn't scale the features, differences in ‘'Total charges'' can have more influence on which customers KNN considers similar than differences in features like contract type, which are encoded as 0s and 1s.

The data set has significantly more customers who stayed than customers who left. This gives the model fewer examples of churn to learn from. It also means overall accuracy can look good while the model misses many customers who leave.

The input columns capture only part of the customer experience. The model sees things like tenure, charges, and contract type. It has no direct information about complaints, service outages, or competitors' offers. Those factors could help explain why two customers with similar plans and bills make different decisions.

The same test set to choose K = 11 and report its accuracy of 78.78% was used. Because the K that performed best on that particular group was picked, the score may overestimate how well the model would perform on new customers. To check this, K = 11 should be evaluated on a separate set of customers whose data was kept aside during training and model selection.

A smaller limitation is also the 11 missing values in ‘Total Charges’ that we filled with median values as they are estimates as opposed to real values and hence can skew the data.